

Finility: News Sentiment, VIX, and S&P 500 Returns (2017–2020)

Goal: Test whether financial news sentiment predicts daily S&P 500 returns.



Approach: VADER labeling → Linear SVC sentiment model → daily sentiment index.



Then merged with S&P 500 returns and VIX to analyze relationships.

Data Sources (Aligned to 2017–2020)

1) Financial News Headlines (CNBC, Guardian, Reuters) — Kaggle CSV (~50k rows)



2) S&P 500 Index (^GSPC) — Yahoo Finance API (daily OHLCV)



3) VIX Index (^VIX) — Yahoo Finance API (daily close)

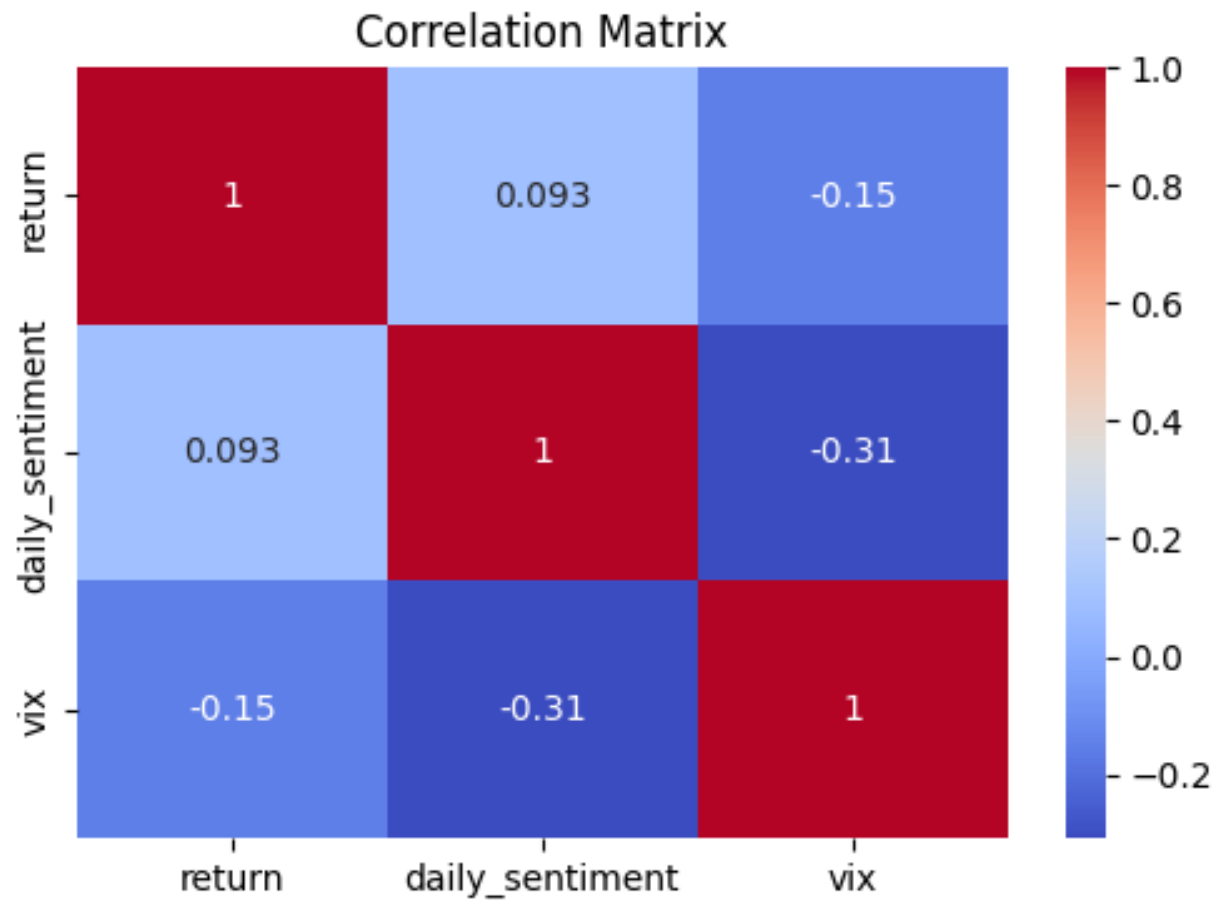


All datasets merged by date; analysis uses window with sentiment available.

Summary Statistics

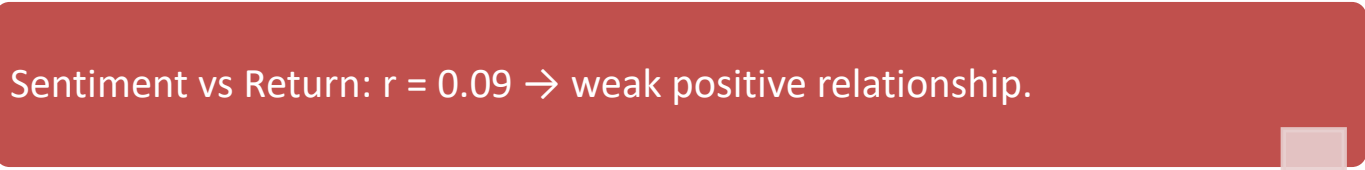
	RETURN	DAILY_SENTIMENT	VIX
count	647.0	647.0	647.0
mean	0.0	0.036	19.412
std	0.015	0.159	10.542
min	-0.12	-1.0	9.15
25%	-0.004	-0.063	13.16
50%	0.001	0.045	15.8
75%	0.007	0.137	20.555
max	0.094	1.0	82.69

Correlation Matrix



Correlation Insights

Sentiment vs Return: $r = 0.09 \rightarrow$ weak positive relationship.

A red rectangular box with rounded corners containing the text 'Sentiment vs Return: r = 0.09 → weak positive relationship.' A light red arrow points downwards from the bottom right corner of the box.

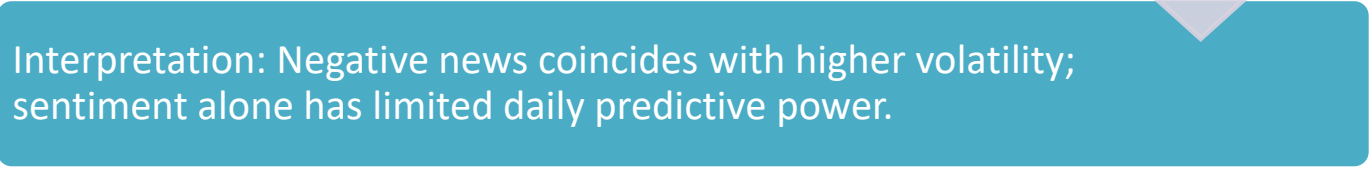
VIX vs Return: $r = -0.15 \rightarrow$ weak negative relationship.

A green rectangular box with rounded corners containing the text 'VIX vs Return: r = -0.15 → weak negative relationship.' A light green arrow points downwards from the bottom right corner of the box.

Sentiment vs VIX: $r = -0.31 \rightarrow$ moderate negative relationship.

A purple rectangular box with rounded corners containing the text 'Sentiment vs VIX: r = -0.31 → moderate negative relationship.' A light purple arrow points downwards from the bottom right corner of the box.

Interpretation: Negative news coincides with higher volatility;
sentiment alone has limited daily predictive power.

A blue rectangular box with rounded corners containing the text 'Interpretation: Negative news coincides with higher volatility; sentiment alone has limited daily predictive power.' A light blue arrow points downwards from the bottom right corner of the box.

Regression Results (OLS)

Model: $\text{return} \sim$
 $\text{daily_sentiment} +$
 $\text{sentiment_lag1} + \text{vix}$

- daily_sentiment : coef +0.0053, $p=0.175$ (not significant)

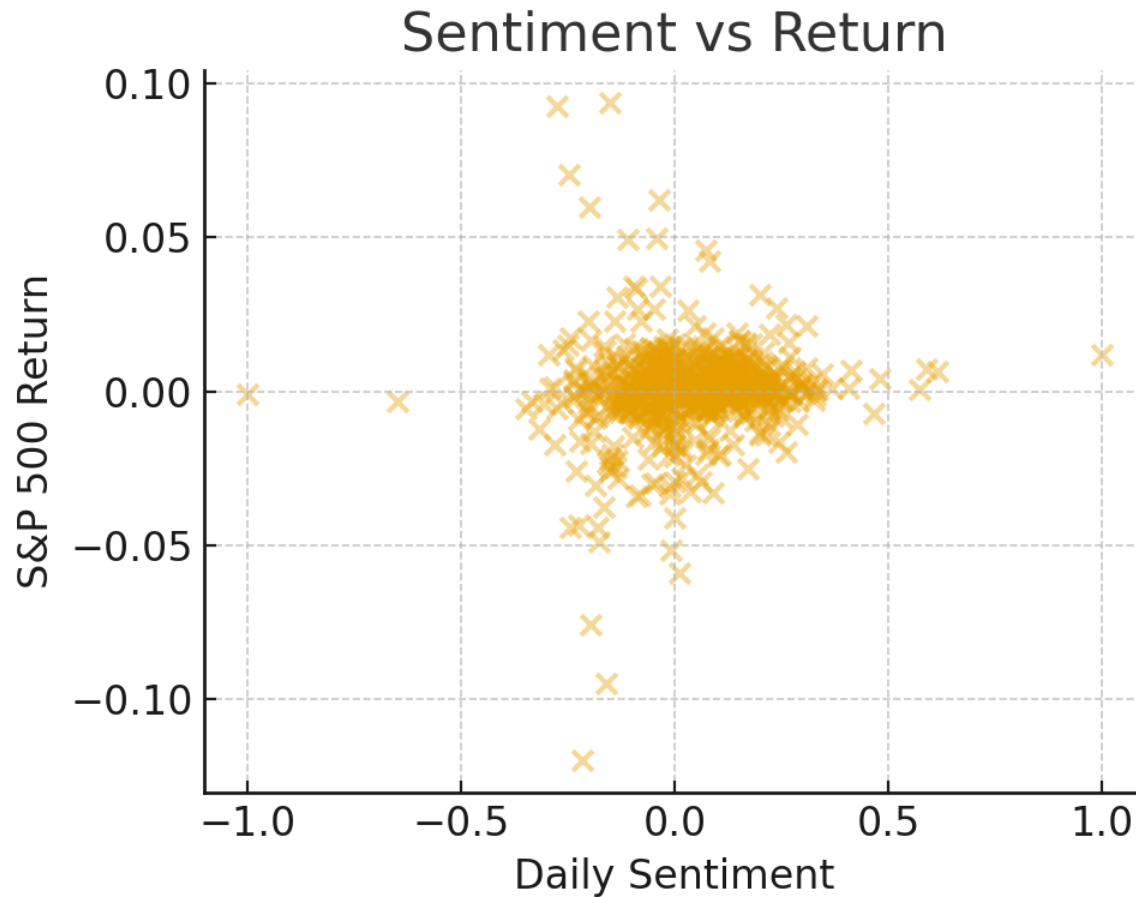
- sentiment_lag1 : coef -0.0048, $p=0.220$ (not significant)

- vix : coef -0.0002, $p=0.001$ (significant)

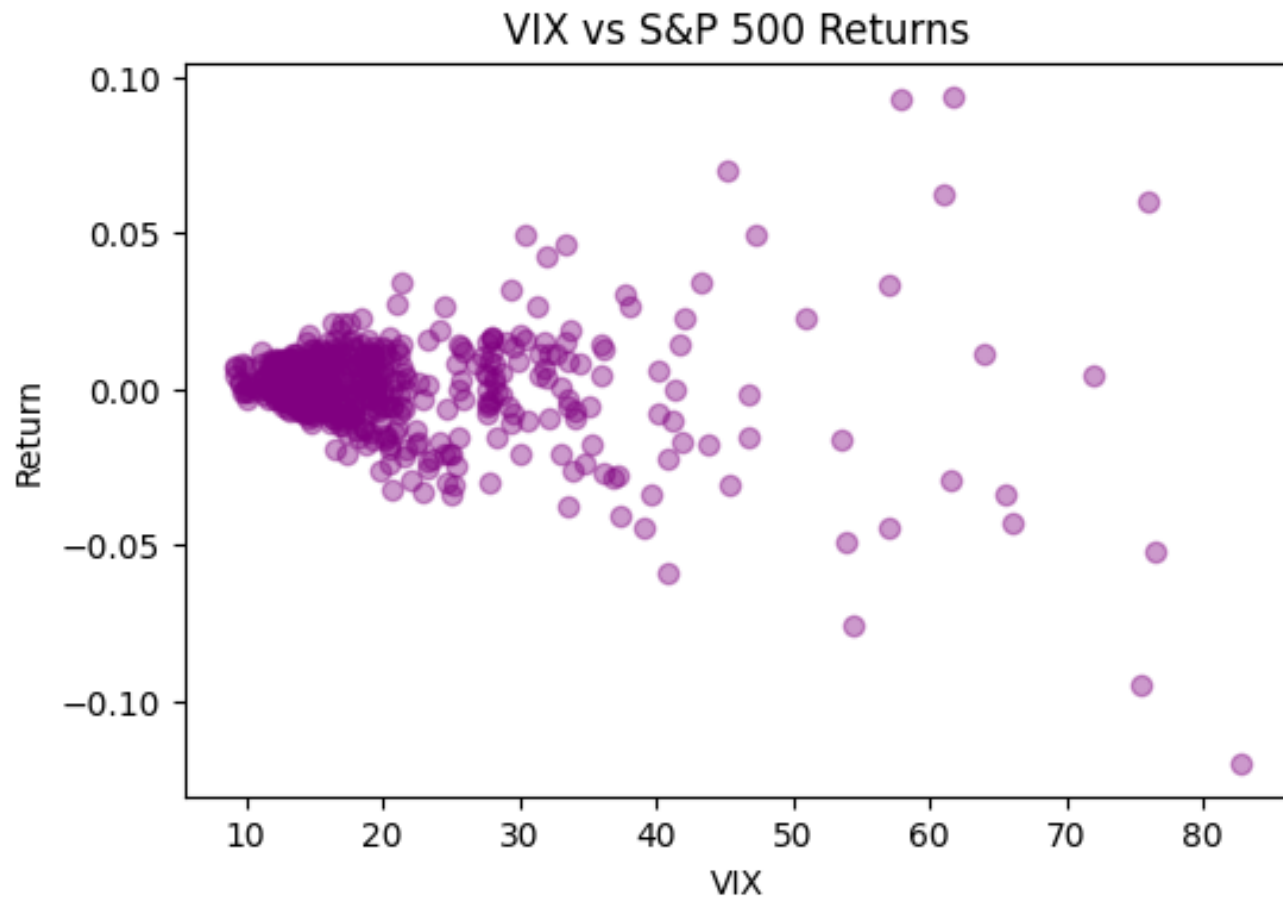
$R^2 = 0.027$

Takeaway: VIX significantly explains returns; sentiment does not at daily frequency.

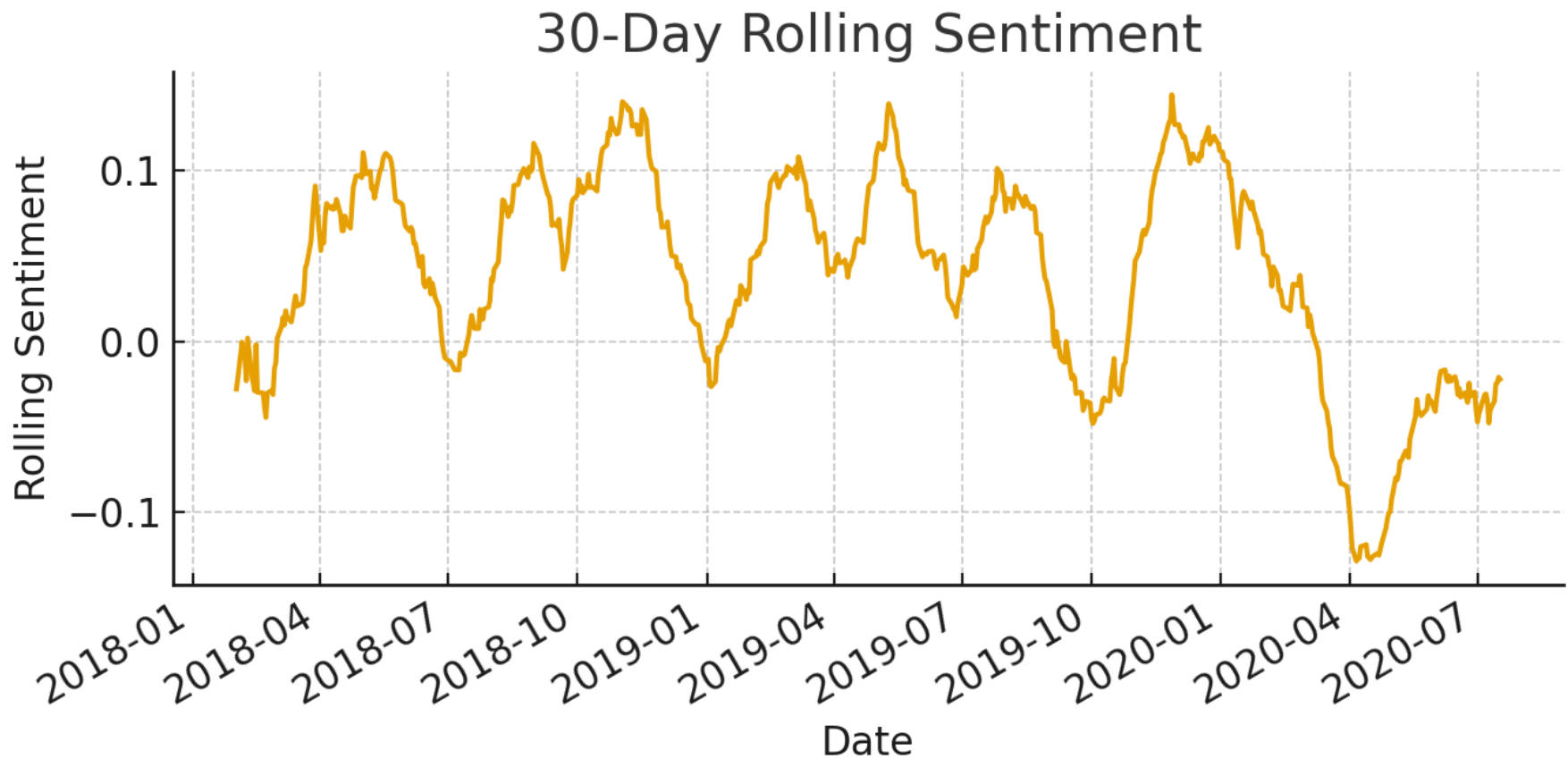
Sentiment vs Return (Scatter)



VIX vs S&P 500 Returns



Rolling Sentiment Trend (30-day)



Challenges



ALIGNING THREE NEWS
SOURCES AND TWO
MARKET SERIES BY DATE.



CLEANING YAHOO
FINANCE CSV EXPORTS
(EXTRA HEADER ROWS).



HANDLING MISSING
SENTIMENT DAYS AND
CREATING LAG FEATURES.



LOW SIGNAL-TO-NOISE
RATIO IN DAILY MARKET
RETURNS.